

Parte seconda
Full English text

SINE COGNITIONE SILEO

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Sundials are no more than the graphical representation, on the Earth, of the apparent motion of the Sun, that moves along the surface of an ethereal celestial sphere. This drawing is obtained by projecting those straight light rays that align the different positions of the Sun with the tip of the gnomon and the shadow on the dial. This immaterial contact between Heaven and Earth is a charming timeless element that makes sundials, even today, a natural meeting place for art, philosophy, geometry, geography, physics and astronomy.

These precious artistic objects, often located in monumental architectures, are the memory of an ancient and valuable knowledge of centuries, when humans regulated their own activities in close connection with the natural and astronomical time sequences, during the day and the whole year. Measuring the time by the Sun and observing its eternal cycles, conditioned for centuries the daily habits of communities, and also rites and liturgies of all religions, based on the periodic repetitiveness of celestial phenomena.

Sundials are the graphic evidence, on the Earth, of some complex astronomical events, always in progress across the sky, at infinite distances from us, and so characterized by the convergence of various doctrines, from mathematics to physics, geography, cartography, geometry art, testifying to the fervent cultural climate has arisen in Europe between the Fifteenth and Nineteenth centuries.

In this rich intellectual scenario, Italy and the Campania region in particular, boast the realization of several sundials of great artistic and scientific value, inadequately valorized for the progressive dispersion of this ancient culture due to the advent

of modern watches. Between the negligence and the inability to carry out adequate restorations, Neapolitan sundials constitute a cultural heritage to be saved with urgent actions based on the dissemination of their value that, in the “city of the sun”, could trigger interesting processes of teaching and tourism disclosure.

Some of those sundials are in a clear state of degradation: very often the time lines have been lost, as in the vertical example in the Floridiana (fig. 1, p. X) and in the Certosa di San Martino (whose orostyles are both still fixed in the facade), while others have suffered inappropriate restoration works that have altered the system for measuring the time. In addition to the ancient examples, in Naples there are some interesting contemporaries sundials, not adequately exploited, as the analemmatic one in front of the railway station of Mergellina and the Tower of the Time, by Massimo Pica Ciamarra, in Piazzale Tecchio (scientific project of Edgardo Filippone).

A good enhancement project about sundials must necessarily start from the knowledge, in order to achieve conservation, improvement and promotion, so laying the foundations for their subsequent use.

The current study has thus identified a methodology able to establish a dialogue between different fields of knowledge (restoration, geometry, new technology, history of architecture, astronomy and gnomonic) with the main purpose of triggering the disclosure, the communication and the consequent use of these singular artistic goods, very often abandoned. The work we present formulates a restoration project of the three ancient sundials in the

Carthusian monastery of San Martino, in Naples, based on the most modern digital technologies.

In the Neapolitan monastery it is possible, even today, to observe three very valuable sundials.

The first, and the most famous one, is the meridian line which is located in the Prior Quarter, built in 1771 by Rocco Bovi: it's a unique and complicated gnomonic example by virtue of its precious artistic bronze bands, on which there are indicated a large amount of astronomical, geometrical and geographical informations. It is placed in a prestigious architecture, which is so closely connected with the meridian line to be considered an integral part of the gnomonic project (fig. 2, p. XI).

The meridian sundial is now well preserved inside the ex library of the Prior, but equally poorly appreciated: despite his immeasurable gnomonic value, which places it among the most complex sundials ever made in Campania, does not exist an opportune exhibition as to make it communicable, and therefore appreciable. The architecture of the room actively collaborates with the meridian line hosted in its interior: the bronze bands are set in a beautiful terracotta tiled floor in *riggole*, by the artist Leonardo Chiaiese, that represent correctly, on the horizontal plane, the projection of the tropics curves and the straight line of the equator, as well as a singular disposition of constellations of both hemispheres, which appears as derived from the ancient representations of the cosmos, including the Farnese Atlas globe.

The exterior windowsill, in which is located in the gnomonic hole, also has a peculiar splay, totally absent in all the other windows, in order to accommodate the entry of sunlight.

A superelevation made in the same Charterhouse during the nineteenth century made it impossible for the light to reach the gnomonic hole during the winter, when the height of the Sun on the celestial equator is lower, with the unpleasant result that, just at the date of the winter solstice, clearly drawn by a finely engraved ellipse in central band, the Sun cannot project its rays on the meridian line. Our survey also revealed that the gnomonic hole, which is in the wooden platband of the entrance door, needs a more appropriate placement to correct some height changes, due to restoration works that have been done over the centuries.

Our research has therefore approached the project of re-functioning the sundial through a system of mirrors, placed on the roof of the building, just in

front of the library, in order to direct again solar rays inside the gnomonic hole during those months in which the height of the Sun is not high enough. The second sundial is located in the big cloister, but the time lines have been erased since the first years of the twentieth century; the dial has been banally plastered although it is still present a perpendicular style fixed in the wall, that casts its shadow without measuring the time anymore (fig. 3, p. XII). From documentary and iconographic sources it has been possible to us to reconstruct the lost time lines and therefore to propose its restoration.

The third sundial present in the Charterhouse is a unique example of multiple dial, carved in a marble block, which presents the most ancient systems of time-keeping, like the Babylonian and the Italian one (fig. 4, p. XII).

Poorly appreciated for the lack of an opportune exhibition in the same museum that hosts it, the clock no longer performs its function for the total absence of the gnomons: we propose to put again on it the lost styles, according to the dimensions we determined, for each quadrant, by virtue of a geometric method.

We intend, therefore, to bring back to light an ancient knowledge, now under dispersion, starting from its best examples, for gnomonic complexity and placement, because they are situated in one of the most important Italian museum, able to attract significant group of tourists and so, able to act a large cultural diffusion. Reactivating the lost function of measuring time, but mainly creating a new, and more appropriate, representation model can still now communicate the complexity of astronomical phenomena connected to sundials.

The entire research was conducted with a dual method, weaving astronomy and gnomonic with the geometry, for the construction of a 3D model to verify and plan sundials restoration, for which, however, have been indispensable the historical and documentary researches that we conducted in the most Italian archives, thanks to which we could interpret the real intentions of the author, testified directly by the the original manuscripts we found in Naples, by the newspapers of the time, but also by the correspondence that Rocco Bovi maintained with eminent figures of his cultural scene, containing very useful descriptions.

For example, the geographical locations engraved along the side bands of the sundial, each showing

an angular value in correspondence, have been correctly identified, despite the initial ambiguity due to their ancient nomenclature, by virtue of the explanation written by Bovi himself who, in the manuscript – totally analyzed in the third chapter -, describes them as places with the same midday of Naples, which therefore are placed along the same terrestrial meridian.

The analysis of these sundials, therefore, has been characterized by a purely geometric cut, graphically checking all the possible alignments with the Sun, during the entire annual cycle, without any reference to abstract mathematics or trigonometric formulas; indispensable tool of investigation and verification, drawing assumes in gnomonic studies the indispensable role of a communication tool, thanks to its better efficacy in popularization.

The installation of a permanent exhibition in the Museum can help to arrange appropriate thematic visits, with the support of exhibition panels and eventual multimedia content for an appealing communication of the artistic qualities of the sundials but, at the same time, rigorous in the scientific description of the astronomical phenomena involved. We hope, in this way, that the research and the studies about sundials will return in the disciplinary field of the geometry, from which they originated, for centuries, as demonstrated by the major ancient treatise writers, who frequently dedicated several chapters to this discipline, before it became a strongly confined theme in a restricted range of fans and experts.

A large and central chapter dedicates a lot of pages to Rocco Bovi, important scientist in the cultural scene of the eighteenth century, describing the multidisciplinary nature of his training, intertwined with the major scientific personalities active

in European cultural scene of his time.

Some of the documents discovered in our archival researches resulted as nodal elements for the geometric interpretation of some complex gnomonic details provided by the sundial. Other papers have instead been useful to refute previous interpretations of critics and scholars, sometimes too general and some other times misleading the scientific consistency of the entire production of Rocco Bovi, to whom we owe some of the greatest artistic masterpieces in the field of gnomonic. Sundials refunctionalisation in the Carthusian monastery of San Martino can be an excellent opportunity to build a network of cultural routes otherwise denied to citizenship and visitors, to discover the rich heritage of sundials in Naples and, subsequently, in the region. The goals of our study can also be replicated throughout the country and, for the particular role in the management and enhancement of cultural heritage, proposed as a “reference model” at the international level.

This book is the result of a long research, began during the Thesis in Architecture of Roberta Murolo, for which I took advantage of the correlation, as regards for astronomical matters, of the Ph.D. Luca Santoro. The common passion for the gnomonic and the awareness of its highly needed re-actualization, due to innovative researches and dissemination events, has led us to discover a rich cultural scene, multidisciplinary, in which these precious examples have been made, driving us to continue the investigation through documentary sources often unpublished.

With our gaze directed toward heaven, and surveying the traces of the Sun on the Earth, with the help of the geometry, even today we travel searching other forgotten time lines.

Chapter I

*THE LINE THAT MEASURE THE INFINITY.
DARK ROOM MERIDIAN LINE OF ROCCO BOVI IN
SAN MARTINO*

Roberta Murolo

1.1 The dark room sundial

Meridian line is a particular kind of sundial, which plot the course of the noon-day Sun as it moves along a line marked on the floor, so working as an eternal calendar where it's possible to read the dates, rather than the hours. These meridian lines are well defined by two kinds of elements tied between them by a close correspondence: the sundial line, which indicates the north-south direction, and the gnomonic hole, from which the rays of light cast, on the previous nominated sundial line, a little shining elliptic shape. It's easy to find these kinds of sundials in close environments, such as Convents, Carthusian monastery, churches, or astronomic observatory. One of the most important dark room sundial is actually located at the Archeological Museum in Naples: 27 mt long, it's extended all around the Great Hall, a room whose name depends on her. The majestic hall would accommodate, after have been transformed in Bourbon Royal Museum, the Farnesiana Library. In the north-west area of the building, according to Giuseppe Casella's will, should have been built an astronomic observatory, not constructed anymore due to financial issues. Because of its calendar function and astronomical reading, this precious gnomonic tool, was used both in religious and scientific environment. Another example is Gian Domenico Cassini's sundial. In 1655, in consequence of the extension works of San Petronio Basil in Bologna, he projected again Ignazio Danti's sundial², previously designed in 1575. On the occasion of restoration work at the Basil in 1776, Eustachio Zanotti printed and integrated

some Cassini's works in his book titled *La Meridiana del tempio di San Petronio rinnovata l'anno 1776*. In the preface it's possible to read:

“Non è duopo rammemorare con qual particolare affezione di chi presiede gli Studi in questa Università sia sempre stata riguardata la scienza astronomica, bastando a persuadere la sola testimonianza, che a noi reca la grandiosa linea meridiana del Tempio di San Petronio costrutta nel secolo scorso dal celebre Cassini. Questa linea avendo poi sofferto quei danni che apporta il tempio a tutte le opere umane, era divenuta incapace di più servire a quelle sottili ricerche attorno al movimento del Sole”³.

The text is like a sort of diary, including construction phases and all the analysis, corrections and gauging taken of the sundial. The meridian line also allows the knowledge of:

“i gradi di distanza al Zenith, le loro tangenti, I segni dello zodiaco, le ore di cui si compongono le notti, i secondi e i terzi della circonferenza della Terra, e la larghezza dell'immagine del sole in estate”⁴.

Moreover it allowed many high precision measurements: Cassini defined it as an “heliometer”, due to the possibility to measure Sun apparent diameter, whose dimension changes during the year, as a consequence of the eccentricity of Earth's orbit⁵. Another meridian line which, some years later, will compete with the one in Bologna was built in

1702 by Francesco Bianchini, into the Church of Santa Maria degli Angeli e dei Martiri in Rome. In 1700 Pope Clemente XI set up a committee, chaired by Cardinal Norris and Bianchini as secretary. The committee had to verify if the Gregorian Calendar Reform was still valid and had to discuss about the Easter date. For this reason, Bianchini was immediately asked to realize a meridian line which marked the day of the vernal (spring) equinox⁶. The effective work of Bianchini's sundial was also studied by the scientist Anders Celsius, when he came to Rome in 1734⁷. Nowadays, it remains one of the most fascinating and analyzed sundial in Italy. Totally different from the other, the sundial of San Martino Charterhouse in Naples has been unknown for long time. In fact there are no many documents about it. That's why our researches focused on Rocco Bovi, his creator.

Astronomer, he was born in Scilla (1734-1831), and he went to Naples to teach Experimental Physics at the Real Collegio del Salvatore and he became professor of Mathematics in Salerno. He collaborated with the geographer Giovanni Antonio Rizzi Zannoni, while he was preparing the big map of the Kingdom of the Two Sicilies. Because of the earthquake in South Calabria, in 1783 he came back to Scilla, where his family lived and where he definitely settled, becoming here Major. His life, who left as heritage the sundial of Abbey in Cava de' Tirreni too⁸, will be carefully analyzed in the second chapter of this book. Of fundamental importance was the discovery of a signed manuscript in the drawings section of National Archive in Naples. Written by Rocco Bovi in 1769, it was entitled *Spiegazioni del presente disegno*. This document, that allowed us to get more about the information around the Neapolitan meridian line, will be analyzed in the second chapter. Thanks to this documents it's possible definitely to attribute the paternity of the Neapolitan sundial to Rocco Bovi⁹. In Nicola Severino's article, *La linea Meridiana della Certosa di San Martino a Napoli*, is published a photo where it's possible to see Bovi's signature, engraved in the ellipse that strikes the winter solstice along the sundial¹⁰. Unfortunately, because of the engraving fading, it's almost impossible to recognize the signature nowadays. Moreover, we have to underline how different from the other sundials is the San Martino meridian line because of the great amount of information that people can read on it. Probably that's why there are so

few studies about it: it's smaller than the others (it's 7 mt and 85 cm long), but it's extremely complex and its interpretation is not so easy.

1.2 The Prior's quarter library hall

People can visit the Prior's quarter library hall since 1995¹¹, consequently to some restoration works. The hall hosted in the past a huge number of artworks. Most of them have been lost during some historical events, such as the suppression of the Carthusian order, when all the paintings have been moved to the Royal Museum¹². After the earthquake in Irpinia, some rooms in the Prior's quarter have been restored, even though the area remained closed to visitors¹³; in particular, Guido Donatone writes in his book entitled *Pavimenti e rivestimenti maiolicati in Campania*, that the library hall was used as a storehouse¹⁴. The renaissance of the whole area, whose opening interested also the ex-library, was in 1995¹⁵ thanks to restorations works directed by Adele Pezzullo and held by the superintendent Nicola Spinosa, but Rocco Bovi's sundial beauty is still overshadowed¹⁶. The ex-library is identified as Room n°33 in the museum; there, it's possible to notice an informative panel indicating the name of the hall, and the hypothetical reconstruction of the ancient furniture, constituted by fixed shelves along the walls. No mention was given to the precious gnomonic good that so lays on the ground such a simple decorative elements (fig. 6, p. 6).

The Prior's library is a very little rectangular hall but architectonically and artistically very valuable. On the two cylindrical surfaces of the pavilion vault, people can admire two paintings from the first half of 1700: San Martino in Gloria and San Bruno (receiving the rule of the order from the Virgin and the Child), both of them attributed to Crescenzo Gamba¹⁷. Once you get in the hall, eye is caught by the floor, painted by Leonardo Chiaiese¹⁸, leader of a very important family that, together with the coeval Massa, played a central role in majolica handicraft in Campania¹⁹. Born in Abruzzo, Leonardo Chiaiese became one of the most important artisan in Naples. His personal style is far from the Rococo style that characterized his family's work and, in part, also his brother's work. In 1743²⁰ Leonardo worked with him at the realization of the floor into the Suor Orsola Benin-

casa church. In reason of that, he was called in 1744 by Antonio Vaccaro to create the floor of the library of Tarsia²¹.

His original inspiration could be noticed thanks to the majestic floor into the church of San Michele at Anacapri, built by Domenico Vaccaro, which is a huge image representing the Expulsion from Eden. To create the floor of San Martino library, Bovi so called Leonardo Chiaiese and asked him for a rigorous work (fig. 7, p.8). In fact, we can notice the representation of some constellations with the curves of both the celestial tropics, Tropic of Cancer and Capricorn. They are represented on the floor, close to the meridian line, in correspondence of the dates of the winter and summer solstices. Equator line is just a short segment, corresponding to the vernal and autumn equinoxes. A deep geometric analysis will be done in paragraph 1.4 of this chapter: it brought out the consideration of the existence of a precious link between the sundial and the floor. Moreover it's possible to notice the picture of some planets, such as Jupiter, Mars, Venus, Saturn. According to *Magazzino Toscano*, this is a representation of "Planets aberrations"²². On the wooden surface of the window lintel it's possible to notice a square metal plate with a Sun, with a human face, whose mouth hosts a small gnomonic hole, of about 11 mm of diameter.

The gnomonic hole is a circular hole across which the sun rays enter and project a shining ellipse along the sundial line (fig. 1, p. 1). Actually, the gnomonic hole height is 3,45 mt from the floor. It's very important to have the right height to establish the well functioning of the meridian line. In fact, some specific evaluations we'll analyze in paragraph 1.5 of this chapter, could be influenced by even only 1 cm of difference.

Two windroses are close to the sundial. Bovi describes them in his manuscript, dated 1769, as:

"ornamenti al menzionato disegno, fra i quali vi sono I due rombi dei venti; uno che fa vedere la direzione dei trentadue venti sull'Orizzonte, l'altro che mostra il vero Polo Boreale ed Australe: serve il presente di norma alle Bussole nautiche, [...]"²³.

The west windrose originally contained a compass, as Rocco Bovi describe and Antonio Coppola mentions in his book²⁴: the circular glass at the center of the windrose hide a box in which, proba-

bly, there was a magnetic needle. Along its edge it's possible to notice the degree of the compass. There are also the four Cardinal points, represented by particular symbols: the Sun indicates the South, while West is symbolized by a red heart with a flame, as the Jesuits one. This order was born in Spain, the land of Ferdinando IV. Then we noticed the lily symbol, also called *Fleur de Lys* introduced, during the maritime republics era, to indicate the north in the compasses. This symbol is also visible on the coat of arms of the Bourbon Royal House, and on the Kingdom of Two Sicilies flag. Finally, the Templar rood indicating the east. The eastern windrose which indicates the continents is nowadays rotated of 90° with respect to the original position.

1.1 Rocco Bovi's meridian line

Laying on Leonardo Chiaiese's majolica majestic floor, Bovi's meridian line is composed of three bronze bands, on which are engraved a sequence of astronomic, geometric and calendar information, surrounded by white marble slab and by decorated and glazing tiles.

The first detail that cannot escape to those who, endowed with sensibility, enters in a place of such artistic value is the formal accuracy of each single detail. In fact, while the side bands are designed as two slender pillars, the central one begins with a rounded tip. Based on the stone step located under the big window of the hall, there are three capital letter, CAR. They symbolize the Carthusian order and the abbreviations of the terms that Nicola Severino muddled with Rocco Bovi's ones²⁵. These letters are, indeed, located on the windows of the Charterhouse and in other different points of the museum. A little bit further ahead, there is the inscription PERPENDICULI PARTES DECEN MILLESIMI (FIG. 8, P. 10). It's referred to the tychonic or geodetic scale below²⁶. It's mentioned in Rocco Bovi's manuscript:

" finalmente si osserva la varia degradazione del sole con le tangenti prese a porzione della scala geodetica..."²⁷.

It allowed Bovi to draw, on the middle bronze band, the value of tangents, which will be described in detail in the following passages.

As also the *Magazzino Toscano*²⁸ argues, talking about this sundial, the tyconic scale has been divided into one hundred equals parts, one of them being the thousandth portion of the radius drawn from the center of the circle located in the northern part of the graphic scale. It ends with the gnomonic hole, located on the lintel of the window. From the little circle represented in the low of the geodetic scale, truly begins the meridian line:

“... Linea Meridiana che principia dal centro del Raggio, che viene ad essere sotto il Zenith, o sia sotto il vero grado di Latitudine, [...], e che termina là dall'Equatore, cioè nell'altro Tropico Australe, o il Solstizio Iemale”²⁹.

In the circle, it's possible to read the sentence *centru gnomonis*. This circle is, in fact, the orthogonal projection, on the horizontal plane of the sundial, of the gnomonic hole located on the lintel of the window. Defined as gnomonic center, this point defines a zero position from which are possible all the angular and trigonometric tangents measurements, that allow the observer to know the solar declination³⁰. On the side of the true sundial line, identified as the central axis, it's possible to notice two areas where some values are engraved. They give out some geometric information, not easy visible in other sundials, even though they're more famous than the one studied here. Before explaining the meaning of this values, it's important to underline that the work was realized by a professor, an astronomer, a scientist who used all of his knowledge to face this kind of study previously faced by Bianchini and Cassini. They went “technician” and experts in Architecture, as Bovi himself writes down in his work:

“ed il tutto fatto dal qui sottoscritto Autore a penna senza che mai avesse appreso regola di disegno od altro appartenente alla Civile Architettura...”³¹.

As already exposed at the beginnings of this chapter, the rays of light entering into the gnomonic hole, each parallel to the other, changing direction during the year. These inclinations are well defined by two values: the former, regarding the angle degrees created with the vertical line passing from the zenith (otherwise known as the gnomonic hole)

by the central ray of light; the latter, corresponds to the trigonometric tangents, of these degree, expressed in decimillesimal. Basically, we have to imagine that the height of the hole center and the distance of the shining projected ellipse from the *centru gnomonis* are both catheti of a right-angled triangle, in which the hypotenuse is the sunray that everyday enter into the hall. The height of the hole always is the same, while the only changing value is the distance of the solar ellipse and, so, the tilted hypotenuse, more horizontal during the winter, when the sun is lower. That's why the spaces where the numbers are engraved, enlarge themselves as well as there's the passage from the summer solstice (represented by the first, and also the littlest among all, ellipse) to the winter solstice (represented by the prolonged one at the end of the sundial). It's possible to notice that from the tyconic scale begins a degree where the numeric values engraved represent the sundial inclination angle degree regarding zenith vertical line. These are expressed in spaces of sixty minutes each, also divided in some other spaces of five minutes. Referring to these degree, Antonio Coppola talks about:

“semiluce diurna espressa in ore e minuti con possibilità di determinare per ogni giorno l'ora dell'alba e del tramonto, l'altezza e la declinazione del sole”³².

The explanation is missing and it's not clearly explained why these degree are called “hour” and, according to what kind of relationship it's possible them to mark the dawn and the sunset, if the sundial lines is active only at midday. Thanks to the words we can find in winter solstice ellipse, which defines a real title for the sundial³³, we know that it reads an arc of the celestial sphere going from the zenith (zero degree) to 64°30' (fig. 13, p. 15). The trigonometric tangent of the aforesaid degrees are equal to the numeric values that are engraved on the sundial line. The segment of meridian line that goes from the *centrum gnomonis* to the first engraved ellipse, in correspondence of the summer solstice, is not active. According to the definition of solar declination and referring to the latitude of the considered place³⁴, this line is the projection an arc of the celestial sphere (from zenith to the considered local latitude) not crossed by the Sun. It's possible to notice the representation, in scale, of

some planets, each of them well represented according to the knowledge of the era, but they're not recognizable at first glance (fig. 9, p. 11). They're listed by Rocco Bovi:

“second dicesi Planetaria: poichè in essa Linea Meridiana sonovi delineati... i Pianeti, I quali hanno la loro aberrazione nell'Eclittica...; scolpiti questi alli Tropici della Linea Meridiana, in quell metallo, al quale credesi che presieda quel Pianeta: come sarebbe... [...] il Sole, di metallo dorato, la Luna d'altro metallo argentato; Marte di ferro; Venere, di rame; Saturno di piombo ecc.”³⁵.

If we watch them closely, it's possible to notice that the material surface has been lost in the last three planets, but it's still possible to distinguish: Saturn, represented by its rings, Mars (its symbol it's very clear) and Jupiter, represented as the biggest among all, with a tin surface, according to the alchemical knowledge of the era. This configuration represented the Planet aberration on the ecliptic³⁶. The position of the planet along the line is not accidental but is linked to the degree we've spoken before, because the aberration is going to be calculated through the longitude³⁷, from South: Mercury, the Moon, Venus. Along the sundial line there are seven ellipses engraved with the image of the Sun, and inside them we can find a large number of data. From the southern side of the meridian line, it's possible to read some Latin words engraved on the marble that surrounds the bronze band: AESTIVUM SOLSTITIUM. The ellipse is that projected by the sunray on June 21st, when the summer solstice begins and when the astrological constellations of Cancer begins³⁸. In his interior there are two indications: SIGNA ZODIACI ASCENDENTIA, SIGNA ZODIACI DISCENDENTIA and AESTIVUM SOLSTITIU CANCRI TROPICUM JUNI XXII; next to them the Cancer zodiacal symbol there is sort of small village. There are two possible hypothesis about the ascending and descending star signs: the descending constellations could be identified as the place where the Sun rise and so, not considering the orientation of the room. for instance, when the summer solstice begins, the Sun rise in the Cancer (fig. 20, p. 18). At the same time, Capricorn constellation, on the other side of the ellipse, descends,

or rather sets. On the contrary, it's possible to interpret the two writings on the opposite sides: the former is west side oriented, so representing the ascending constellation that are visible at the sunset, while the latter is at east oriented, indicating the descending constellations that won't be visible anymore because the Sun is rising. Inside each engraved ellipses are indicated the date, together with the constellation of the day. On the winter side of the meridian line, we can find one of the most important point, that's to say the vernal equinox that coincides with the autumn one (fig. 10, p.13). As like as the summer solstice, also for this date there are these marble words engraved on the west side: AEQUINOCTIUM AUTUMNALI (September 22nd) and AEQUINOCTIUM VERNALE (March 20th) with both the signs of Libra and Aries. Moving forward, we'll find the Sun in Scorpio and on the opposite site, the constellation of Pisces. Passing through November 21st in the Sagittarius constellation and, on the opposite side, in the Aquarius (January 20th), there is the longest shining projection of the whole year: the winter solstice. Inside the big ellipse is engraved 1772, that is to say the year when probably was completed the sundial (fi. 11, p.14). Inside we can also read TROPICUS CAPRICORNI, whose constellation strikes the passage of the Sun on December 21st. Moreover, there's a visage's representation, that Nicola Severino³⁹ suppose to be that of the author itself. It's possible to notice the picture of a sailing-boat: we hypothesize that it could represents Bovi's long journey from Scilla to Naples⁴⁰. But this sailing-boat could also represent Ferdinando IV's naval fleet because to him is dedicated the sundial⁴¹. The edge of the ellipse is shaped by a sentence: MERIDIAN SEMITA AD FIXAS A VERTICE AD GR. 64 & MIN. 30 IN SEP-TERNTR BRUMALE SOLSTITIUM HABENS A DIE XXI MENSIS DECEMBRIS.

As already said, this engraving explains that, on the meridian line, it's possible to read just the arc of celestial sphere, measured from the Zenit (64°30'), along the meridian of Naples. Remaining on central bronze band, it's possible to notice, next to the trigonometric tangents, some little words referring to some geographical places. From the winter ellipse previously mentioned, it's possible to see, in correspondence of November 21st (or January 22nd) the name of Frio Golfo, represented by two ships, then Rio Vieve, represented

by the river, and Rio Saele, Angra de Negro, Ance du Nigir, Masoula, Zaire Rio, Populi Bramas Dacti, Corsico, Leeunde Rio, Populi Amecsini, Biafare, Bito Regnum, Secmor, Digir, Agades, Vieve Tripoli. They're rivers, villages, countries located along the meridian of Naples: each of them is linked to a latitude value, calculated as a difference between the local one and Naples⁴². The last detail, which is also one of the most fascinating, is the representation of constellations divided in group of stars, passing for the meridian of Naples. Rocco Bovi also created a calendar, whose structure reminds to the Gregorian kind and, as argued on *Magazzino Toscano*, and it's calculated up to 370043 d. C. As already said, this engraving explains that, on the meridian line, it's possible to read just the arc of celestial sphere, measured from the Zenit (64°30'), along the meridian of Naples. Remaining on central bronze band, it's possible to notice, next to the trigonometric tangents, some little words referring to some geographical places. From the winter ellipse previously mentioned, it's possible to see, in correspondence of November 21st (or January 22nd) the name of Frio Golfo, represented by two ships, then Rio Vieve, represented by the river, and Rio Saele, Angra de Negro, Ance du Nigir, Masoula, Zaire Rio, Populi Bramas Dacti, Corsico, Leeunde Rio, Populi Amecsini, Biafare, Bito Regnum, Secmor, Digir, Agades, Vieve Tripoli (fig. 25, p. 20). They're rivers, villages, countries located along the meridian of Naples: each of them is linked to a latitude value, calculated as a difference between the local one and Naples. The last detail, which is also one of the most fascinating, is the representation of constellations divided in group of stars, passing for the meridian of Naples. Rocco Bovi also created a calendar, whose structure reminds to the Gregorian kind and, as argued on *Magazzino Toscano*, and it's calculated up to 370043 d. C. Calendar data are in correspondence to the AESTIVUM SOLSTITIUM (fig. 22, p. 19). It's possible to read, from the external side: TANGENTIS DISTANTIARUM SOLIS A PROXIMO AEQUINOCTIO. Practically, it's indicated once again information about the sunray inclination, as like as it happens on the central band but, this time, day by day. We find, in sequence, other words like DIES MENSIIUM and CYCLUS SOLIS SIVE EPACTA. The *Magazzino Toscano* argues:

“ in terzo luogo ben distintamente calcolato vedesi il Cielo Solare, o sia Epatta, che utilmente serve per ritrovare I Pleniluni, quando accader deggiono dell'anno 1773 fino al 2700”⁴⁴.

Knowing when the full moon will be, it allowed to know the exact date of Easter day. According to the Christian rule introduced by the Gregorian reform, Easter day would be celebrated the Sunday after the first full moon, after the vernal equinox. The epact⁴⁵ allows to link the age of the moon to the solar year, in such a way that it depends no more from a directed astronomical observation. Because of the weather condition, in fact, it's not always possible to do it in the sky. The annual or daily, as in this case, epact calculation is extremely complex. For more information, I suggest to get a look at the bibliography⁴⁶. In some points there is the representation of stars, indicating the 30th lunation, or the new moon for the year⁴⁷. The engraved LITTERAE DOMINICALES, according to the Gregorian reform, indicates the letter associated, each year, to the Sunday. For instance, in 1771, Sunday was individuated by letter F.

The words ECLIPTICA GRADUS literally means “ecliptic degree”. A meridian line represents, on the horizontal plane, all information about the Sun. In this way, Bovi projected, for each day of year, on the sundial horizontal plane, the longitudinal degree of the ecliptic, that is to say the path that the Sun turns across the celestial sphere during the year. It regards the four points of ecliptic and are defined from the summer solstice, autumn equinox, winter solstice and from the vernal equinox. (fig. 17, p. 17) According to this path, we can have four spaces from 0° to 90°. Element of comparison between Bovi's meridian line and the other more famous ones, is not only huge number of information offered, but also the arc which connects the two lateral bands. This is entirely dedicated to the Easter. We can divide it into three parts: the first dedicated to Easter, that could be celebrated on March; the third to Easter celebrated in April. Between them there's another band where there are engraved all the Planets, as like as they appear on the first segment of the meridian line, the Sun and the Zodiac constellations, from Aries to Pisces. Knowing the epact number that corresponds to the year and the relative dominical letter, we can obtain the date of Easter.

1.2 The floor role and its link to the sundial

Studying Rocco Bovi's sundial we asked ourselves about Chiaiese's floor role. Why are represented only few constellations of both the hemispheres? What exactly are those curved white stripes highlighted in? Is there a correspondence between what is represented on the floor and the directions on the sundial? According to what kind of order are represented the four Planets at north on the meridian line? In our opinion, therefore, it seemed appropriated to assume that Bovi's sundial could not be isolated from the majolica floor in which it lays. Some of those questions have been analyzed thanks to a careful search through documents and a comparison with the analogous historical researches⁴⁸. Other details, after a geometrical analysis, revealed something interesting. While in the second chapter will be faced, in such a detailed way, the problem about the planets and the constellations, this chapter works as an introduction of a study that allows to verify that the whole floor was created as a large sundial whose midday line is the meridian one. First of all it's possible to notice the words on the curves, marked in white, represented in correspondence of the ending parts of the sundial (fig. 26, p. 22). The Southern states: TROPICUS CANCRI, while the northern: Solstitium BRUMALE TROPICU CAPRICORNI. Between them it's possible to notice the word AEQUATOR. In the heliocentric model both the Tropic of Cancer and Capricorn are parallels, the first to north and the second to south, regarding the Equator line and they represent all the places on Earth where the sun is at Zenith just for one day pro year, that is to say, when begin the winter and the summer solstice. Their latitude is $23^{\circ}27'$ north (T. of Cancer) and $23^{\circ}27'$ south (T. of Capricorn). The same happens to the Equator: in its correspondence, sun is at zenit (it means at maximum height) at Midday during the equinox. Day and night, moreover, last the same. Its latitude is 0° . In the geocentric model of the celestial sphere, the tropics represent the parallels marking each declination reached by the sun during his annual motion: on Tropic of Cancer, the sun reaches it on June 21st ($+23^{\circ}27'$); at Tropic of Capricorn the Sun reaches it on December 21st ($-23^{\circ}27'$). The celestial equator is the place of points with no declination and it corresponds to the Equator of Earth's projection on the celestial sphere. The ecliptic,

the path turned by the Sun around the Earth during the year, touches the celestial Equator on March 20th (Vernal Equinox), and on September 21st (Autumn Equinox), while touches the Tropics on June 21st (Summer Solstice) and on December 21st (Winter Solstice). Analyzing the position of those curvilinear sections on the floor regarding the sundial, we can notice that the line of Tropicus Cancrī is exactly juxtaposed on the picture of the sun on June 21st (Summer solstice); the line of Aequator, imagining its projection till the intersection with the sundial line, reaches the eclipse representing both the Vernal and Autumn equinox; finally the Tropicus Capricorni line corresponds to the longest eclipse, marking the Winter Solstice (December 21st). This seems to be moved regarding the referring date posted on the sundial. Maybe this is due to some execution drawbacks of the pavement. Following the geometrical method for the gnomonic study and putting the center of the sphere in the center of the gnomonic hole, it's possible to notice that the cone-shaped surfaces, starting from the gnomon and have the arrangement of the Sun as bases, in the declination of $23^{\circ}27'50''$, intersect the floor in correspondence of the solstice representation previously described, while the projection passing through the equinox curve not only distinguishes a zero point declination, but it also cross the AEQUATOR (fig. 27-28, p. 23). If the sunrays enter in a very far from midday on, for instance, June 21st, we could see the shining allipse of the Sun far from the sundial, eastwards on Tropic of Cancer line. It doesn't happen anyway because the hole, next to a punch in the masonry, allows the entrance only of the midday sun ray.

1.3 The state of preservation and the current problems

Nowadays, tourist that visit the Prior's library can read all the data on the floor and on the meridian line. Thanks to the restoration works, done in 1995⁵¹, the state of preservation it's almost good. Few years ago, as it's possible to notice on the photographs published by Coppola⁵² and Severino⁵³, along the area of the hall, there was a wooden platform that allowed the tourist to watch the sundial closely, but keeping off the floor. Nowadays there is no platform and it's forbidden to cross the hall, in order to avoid the deterioration of the pavement. The terracotta tiles where the constellations are re-

presented, seems to be intact but in some points, the decoration has been removed: it's evident, for instance, on the representation of *Canis Maior*, *Canis Minor* and *Delphinus* constellations, at the west of the sundial. Moreover, *Indus* and *Thuribulum* with *Phoenix* constellations are covered by a carpet. That's why they're going to be damaged, as it happened to *Argo Navis* constellation, corresponding to the west entrance of hall. This constellation is almost entirely deteriorated, because of the carpet rubbing.

After a careful analyze, it's been noticed that some of the tiles have been repaired in some areas. Along the perimeter it's possible to see a double decorative band, a gold one and the other with green, yellow and blue flower motifs. Watching them closely, it's possible to affirm that they're very close and the two fillets, at the same time, are so far and don't belong to a single paint. Next to the door-window and the path next to the *Eagle* constellation, there are some tiles probably set with a logic hard to explain, because they're very different from the other. Is it possible that they belong to Chiaiese's work, probably found in the other museum halls? There's another element that bring us to think that the pavement has been tampered: the wrong position of the windrose which appears rotated 90° regarding the cardinal Points. Concerning the sundial state of preservation, we have to say that is in a perfect one. The information engraved on her are easily readable, but some geographical places and Rocco Bovi's signature faded, like the material surface of planets like *Saturn*, *Mars* and *Jupiter* which disappeared. According to what is explained in the second paragraph, *Jupiter* had to be represented by tin, while *Mars* by iron. Probably the surface has been removed as the years pass by. In consequence of bureaucratic issues, linked to the lecture of the documents preserved by the Superintendence was forbidden to us the access to documents concerning the restoration works done during the years in Prior's quarter. During the large number of surveys, we observed that the sundial is still active but it doesn't work correctly. Practically, it mark the midday correctly but in a different day. The date, in fact, was staggered of about 5 days. For instance: during the survey of May 6th, the center of the projected ellipse corresponded to May 10th-11th. This observation allowed us to hypothesize that, the gnomonic hole could have been moved from its original posi-

tion. It was also confirmed by a report of 1995. From the documents and from the geometrical analysis based on Bovi's calculations, we know that the original height of gnomonic hole center was around 13,5 Neapolitan palms⁵⁴, corresponding to 3,56 m (fig. 33, p. 28).

Moreover, this information was also confirmed by an analytic study concerning the data engraved on the sundial. It means that the value of trigonometric tangent, which is expressed in decimillesimal, is 10000. As regards the angled-triangles we've spoken in paragraph 1.3, the length beginning from the center of the hole to the line where the value is indicated, is equal to the height starting from the vertical line to the center of the hole. This distance has been calculated and its value is 3,56 m.

Nowadays, the height of the gnomonic hole, calculated from the sundial plane, is instead about 3,453 m. Finally, we can say that was made a study according to which the sundial is considered to be still active, but it doesn't work correctly. Another relevant aspect highlighted during the surveys done is that the shining ellipse doesn't appear on the winter season. In fact, even though the hole could be correctly set, the light of the winter Sun won't reach the hall. That's because of a raised super-elevation on the south-west side of Statue courtyard which, during the winter, overshadows the gnomonic hole (fig. 35, p. 29). It has been noticed that this event is verified when the Sun reaches 28°50' of declination (fig. 37, p. 30). It means that the winter part of the meridian line is always overshadowed. It influences the beauty of this sundial. The winter ellipse, in particular that of the winter solstice, is the most fascinating one. The longest dimension taken during the day, when the Sun is lowest and the less tilted radius made it something particularly amazing.

If this sundial will be denied of an event like that, the tourist won't be able to discover not only what's hidden behind the gnomonic tool, but also it's "poetic" expressivity (fig. 39, p. 30).

1.4 Re-functioning project of the sundial of the Charterhouse of San Martino

Angelo Triggianese

The re-functioning project of the sundial in the Charterhouse of San Martino starts from the necessity to make working this fine instrument for measuring time during the 59 days on which the additional storey of the nineteenth century prevents light from passing through the gnomonic hole. As already described in the previous paragraphs, from 22nd November to 19th January, i.e. in the period when the noontide Sun generates rays whose inclination to the plane of the celestial equator is less than $28^{\circ}50'$, the presence in the courtyard of the wall facing north-west, and especially its height, prevents the projection of the bright macula on the bronze strip inside the room, making silent the sundial for a period that lasts about one sixth of the year (fig. 39, p. 32). The problem is not only the amount of time in which the meridian line cannot work but the period involved in this phenomenon, because it happens right on the day of the winter solstice, corresponding to the farthest point from the gnomonic hole along the bronze band. Winter solstice is one of the four main days of the year (two solstices and two equinoxes), when the light macula should coincide with the more elongated ellipse and then in the following days (also not visible) 'goes back' due to the cyclic path that the Sun does in its apparent motion on the celestial sphere. Although there are, therefore, almost two months when, at noon, shadows obscure the gnomonic hole, in metric terms, on the sundial only 1m requires this project because, as just said, it happens in the period in which the reading on the same strip is carried out twice, respectively before and after the winter solstice. The main purpose of sundials in the contemporary age is not only the actual reading of time (because we can use analog and digital precision instruments), but their importance is the inherent scientific value which makes these ancient artifacts actual object of study and interest.

Just because these instruments are often associated to popular activities, or in any case it would be appropriate that there are activities to ensure their knowledge and enjoyment, especially on the days of the solstices and equinoxes, it becomes absolutely essential to plan a re-functioning project of the sundial of the Charterhouse of San Martino so that

the reading of time is possible even in the short, but important, period in which today it remains silent. In fact, the term re-functioning may be improper because, as already described, it is not in discussion the total working of the sundial nor its gnomonic correctness, but rather we should speak about a project to extend its operation, because the aim of this paper is to allow the instrument to be active throughout the year; so the project will be related, therefore, only to the last segment of the sundial. It is considered important to emphasize that the project that we describe here does not impair or affect in any way the regular functioning of the sundial during all the other days of the year (i.e. from January 20th to November 21st), the period during which we do not need any help to the vision of the bright macula. If we consider only the popular character, it would be possible to identify an intervention method that allows at twelve o'clock the genesis of a light projection that perfectly coincides with the ellipses shown on the bronze strip. This objective could be, for example, achieved with a laser sighting system or with a system of adjustable directional lights, whose source can also be placed in a position not belonging to the direction that aligns the point on the sundial with the gnomonic hole and with the Sun, as long as their orientation is calculated in order to fully illuminate the correct position within the room. This would ensure an illusive, albeit not misleading, functioning in the two months taken into account in order to make possible a use related to the informational aspects which make obvious and therefore understandable the astronomical phenomena associated with the reading of the time.

This research, however, wants to be a further goal, i.e. it is essential to put as main assumption the respect for the natural functioning of an instrument that is inextricably linked to the (apparent) motion of the Sun, suggesting, therefore, a method that does not include the use of substitute tools of the real projection of light. We have therefore preferred to find a solution that exploit only the same elements that are the basis of the reading of the time during the remaining period of the year: the Sun's rays. The aim, then, is to make sure that the sun-rays at noon, during the period involved in our observations, follow their natural inclination passing through the gnomonic hole and projecting the bright macula on the bronze strip inside the room. To respect of the historical and artistic value of the

architectonic monument that contains the sundial, we cannot definitely think about bringing the context to the past time (in 1771), demolishing the additional storey that shades the gnomonic hole, so we have identified a tool of project to ensure that sunrays reach into the room, with the correct inclination bypassing the physical obstacle; in other words, it was intended to convey the light rays that obviously arrive in the courtyard, but at an altitude higher than the gnomonic hole, making sure that they can brighten up even the lowest part that remains in shadow. We propose, therefore, to use flat mirrors, exploiting the geometric laws of reflection to divert, according to the requirements and the size of the actual context, the direction of sunrays and then to redirect them towards the aforesaid hole. The mirrors are the tool that has always been used in gnomonics to bring light where there it cannot arrive, in order to build fascinating catoptric sundials, such as to be read under the flat or vaulted roof of enclosed spaces.

“Tutti quegli’Ingegneri, che gustano delle curiosità al vedere le varie sorte di Horologi Solari, che fin’hora sono state inventate, sogliono frà gli altri ammirare quelli, che Riflessi si chiamano, i quali essendo dipinti in luogo, ove non batte mai il Sole, [...] Poiche, oltre l’industria, che pare maravigliosa per descriverli regolatamente in superficie così irregolari, hanno con se il comodo di potere (senza portarsi con la persona al Sole, stando in casa coperto) veder le hore; che anzi si veggono in una stanza chiusa, ove niun altro spiraglio sia, che quello onde passa il raggio solare”.

If the problem we want to solve is the absence of light in the hall of the sundial during the considered period, to solve it we want to use the same methodology used for centuries for the catoptric dials, taking full advantage of the gnomonical tools and, in particular, the geometry and the optics, whose laws describe the phenomenon of reflection of the mirrors. Giuseppe Maria Figatelli in his treatise about sundials of 1675 thus exemplifies:

“Per fondamento dunque di quanto son per provarvi, bisogna supporre una verità à Speculari. (Eucl. secondo il Zamberto Teor. I) tanto certa, e infallibile; che anco à più

rozzi spiriti l’isperienza istessa ne può fare testimonianza; e è: che il riflesso del Sole sempre diametralmente si getta nella parte opposta al di lui raggio con angolo eguale: in tanto, che il riflesso dipende imita, e totalmente siegue il moto del raggio Solare. A quel passo, cha s’alza ò s’abbassa il Sole; à puntino s’alza, e s’abbassa parimente il riflesso: dalche ne siegue; che quanto descrive il raggio del Sole per mezzo della punta dello Stilo ne’ piani ove la di lui ombra ferisce: tanto ad unguem dimostra parimente il di lui riflesso ne’ piani ò Volte opposte, mediante lo Specchio, di dove si spicca, e principia esso riflesso.”

If the perpendicular to the mirror surface belonged to the vertical noon plane of the celestial sphere, whose center coincides with the gnomonic hole, the reflected ray would still belong to the same plane, forming with the aforesaid perpendicular, an angle equal to that of incidence. This allows us to operate on the same vertical surface to which the light rays belong when noon comes during the whole year, not invalidate the proper functioning of the meridian line for measuring time. In fact, unlike canonical catoptric sundials that are represented on the intrados of vaults or ceilings, for which the light macula is generated by the reflected rays after being deviated from the natural course (albeit in a completely symmetrical way), in the case of the Charterhouse of San Martino reflection is used to bring the rays in the same natural direction, after being ‘temporarily’ diverted. (fig. 40, p. 35). The façade containing the gnomonic hole at noon is illuminated only in its highest part: not enough, therefore, only a mirror because we would end up diverting the direction of the sunlight. For this reason it is essential to use two mirrors, one placed on the facade, in a higher position than the gnomonic hole to intercept the light rays that go beyond the additional storey with the aim to redirect them in the opposite direction, and a second one mirror placed on the opposite building (in the same yard) that, following the second reflection, ensures the illumination of the gnomonic hole. Mirrors must be parallel: in this way it is possible to lead, at a lower level, the intercepted light rays, but, at the same time, the ray that enters into the room of the sundial will remain parallel to the ray deflected by the first mirror, thus restoring the

original projection of light. In this way, the project does not alter the gnomonical system but it rather appears as a tool to minimize the external contribution to its operation, returning to the Sun its main role of center of projection, in a very 'natural' way. To locate the mirrors in the correct positions in the courtyard it is appropriate to consider the part of the meridian between the two ellipses engraved on the bronze strip furthest from the gnomonic hole and then covering the period which runs from November 21st to January 21st. If we look at the alignments of these positions with the gnomonic hole and with the Sun, we observe that, on the celestial sphere, we are dealing with an arc on the main meridian subtended by an angle of $3^{\circ}6'$. This configuration allows us to identify unambiguously, after having chosen the location, the two ends of the reflective surface that will be placed on the opposite building. The point A, which allows sunlight to reach the gnomonic hole on the day of the winter solstice, will be high 2,61 m above the horizontal roof of the building, whereas the point related to the days (coincident on the celestial sphere) of November 21st and January 21st (point B) will be located at 3,46 m. Consequently, the rectangular mirror should have the largest side of about 90 cm and the smaller one of only 10 cm. Then, we have to install a second mirror, right on the roof of the building that houses the sundial. Its location and its size derives not only from the distance between the two mirrors (largely dictated by the architectural configuration of the context), but also from the inclination of the reflecting surfaces. We want to tilt the plane of reflection of $4^{\circ}50'$ as to the vertical axis (always respecting the condition of belonging to the plane of noon of the line normal to the surface) to minimize the elevation of the two end points above the horizontal plane of the considered roof. In this way the point A will be at an elevation of 0,50 m while the point B at an elevation of 2,10 m determining, consequently, the longer dimension of the mirror (1,70 m); all the intermediate positions intercepted by light rays in the period of which we are concerned will be included between the two end points. The tilted mirror, even if of an apparently irrelevant angle, determines actually a significant reduction not so much of the distance between the points, but rather of their elevation: designing the mirrors as vertical surfaces, nothing would change for the configuration of the first described item, but for

the second item the elevations of the points would be considerably higher (about 3,00 and 4,70 m, respectively) and it would not be an acceptable solution because it does not consider the need to protect and respect the historical and architectural value of the monumental good. It's essential, then, to verify that the adopted solutions, as project tools of the gnomonical problems, don't alter the architectural status quo.

The structure and mirrors, as described, would be placed in parts of the buildings that are not practicable by visitors and their implementation will not bring any material damage to the involved surfaces, nor require any changes of the status quo. The choice of materials, among other things, is done to underline in a clear and unambiguous way the contemporary root of the design and the character of adding subsequent to the gnomonical project of Rocco Bovi. On the other hand, however, the mirror size to which we have referred, would lead to a strong perceptual impact that, from inside the small yard, may perceive the intervention as a invasion of the monumental context because of the vision of the large and tall mirrors that will be above the involved roofs. Without renouncing the geometric rigour that unequivocally determine, depending on the incontrovertible position of the Sun during its apparent motion, the positions of the points for the reflection, our research has gone into the identification of a solution that does not compromise the architectural context in which we are working. The project, therefore, wants to use two vertical telescopic rods, each placed in the previously identified positions, so that they belong to the virtual plane of the noon; at the ends of the same rods the two mirrors, having the previously mentioned inclination, will be placed. Each rod, made mobile by a mechanical device, may extend or reduce its own length in function of the day of the year, to ensure that the point of end of the rod coincides with the position which is necessary to determine the correct reflection. It will therefore be sufficient a small reflective square surface ($10 \times 10 \text{ cm}^2$) because, day by day, it will be at the needed elevation to direct the light rays towards the gnomonic hole and to generate inside the room the elliptical macula. The structure of the project, therefore, using simple and modern technologies, becomes a dynamic tool programmed to operate only when necessary: during the months when the sundial is naturally enlightened by the Sun, the mirrors

will be retained by telescopic rods in a default position (the lowest one allowed by the size of them), from which they will not generate any reflection; the same principle will also be applied from November 21st to January 21st: only from half an hour before to half an hour later midday, that's to say from 11,30 a.m. to 12,30, the device will operate extending the rods to bring the mirrors in the correct positions, linked to the date, to return after 12,30 in the basic configuration. In this way visitors can see the mirror system only when it is really active, clearly showing themselves as a tool to let the ancient sundial work again. We do not want to deceive observers into believing that the Sun shines directly on the gnomonic hole, but we prefer to make visible the technological apparatus looking unmistakably as a contemporary object, that participate in the larger gnomonical project of which joined. These rods work only for 0.7% of the duration of the entire year (about 60 hours compared to the 8760) and so it's very small their perceptual impact: the mirror placed on the roof of the sundial room it's not visible at all. At the same time, however, we also considered the problem of the elevation at which each of the two elements of reflection needs to be on the days of November 21st and January 21st when, to properly reflect the ray, the greater height is required: placing the rods away from the edges of the buildings, the visibility of the objects from the courtyard would been less evident to disadvantage of the size of the rods in the act of elongation. It was consequently sought a compromise between perceptual impact and sizing of the project to ensure the proper functioning and the desired integration with the context. For the movement system of the telescopic rods it is considered appropriate to perform the required operations by actuators electronically controlled by the *Arduino* platform. This instrument, used to build interactive objects, cyclically executes the commands provided by the programming language to operate the devices connected to it but, at the same time, through some sensors, it is able to receive input from external physical world to adapt itself and respond to stimuli.

Among the infinite alternatives that electronics now makes available to designers we want to use *Arduino* not only for its simple use but also for the close relationship that it can establish with the three-dimensional design software. If the electronic programming often characterized by a very

complex syntax may become an obstacle for architects, who are often responsible, as in this case, of restoration and renovation of ancient sundials, making necessary the intervention of external professionals that, however, doesn't know any gnomonic problem, the ability to overcome these difficulties tightening a direct relationship between the technological functionality and the tools of architectural drawing become an additional resource for the proper design and implementation.

Arduino, in addition to receive commands via the direct writing of computer language, can be managed through the use of plug-in of the same software used to draw the 3D celestial sphere and the sundial. Starting from the virtual three-dimensional model of the context, in fact, we were able to exploit, through *Grasshopper*, a graphical algorithm editor tightly integrated with the modeling tools, the benefits of parametric design that allows to have 3D configurations just depending on the chosen parameters (fig. 44, p. 38). In this way, day by day, starting from the inclination of the light rays and from the alignment between the macula that needs to be generated, the gnomonic hole and the Sun, we can automatically generate the aforesaid geometric configurations in order to identify, in the virtual environment, the position that the two mirrors have to assume (figg. 42-43, p. 37). Setting, from time to time, day and hour as parameters that govern the graphic operations, we can directly display the three-dimensional model in favor of a direct control of the project. The next step, to link the virtual environment to the real world, is the use of another plug-in capable of transcribing graphics operations into computer language, so as to send commands to *Arduino* and, consequently, to manage the movement actuators. The telescopic rods, therefore, installed on the buildings of the Charterhouse of San Martino, depending on the hour and on the day, will remain motionless if the system will not need their intervention or, on the contrary, they will change their extension according to the calculations made by the software, which parametrically change the situations derived from the context, acting directly on the model in three-dimensional virtual environment. In this way the path of our re-functioning plan ends. This path started from scientific studies about the apparent motion of the Sun, i.e. the celestial sphere, to get to manage, with the information obtained as a function of the cyclical nature

of the astronomical phenomena, the mechanical movements that make concrete the new functions. An additional benefit derived from the parametric design/representation is that the described project is not only the solution to the problems of the Charterhouse of San Martino but it can become a simple methodology to be applied in other contexts with similar problems. In fact, changing only the parameters that depend on the environment (such as latitude or geometrical-formal aspects of the existing spatiality) this instrument fits in an immediate way to a possible different site, being able to repeat all the operations necessary to obtain the new geometric spatial configuration.

Note

¹ Cf. chapter 3, paragraph 3.5, of this book.

² Ignazio Danti creates another sundial line in the Specola Vaticana and, before, the Florentian armillary sphere and the gnomon on Santa Maria Novella facade.

³ E. Zanotti, *La Meridiana del Tempio di San Petronio rinnovata l'anno MDCCLXXVI*, Bologna MDCCLXXIX, pp. 3-4.

⁴ G.B. Rampoldi, *Corografia dell'Italia*, Volume Secondo, Milano, MDCCCXXXIII, p. 650.

⁵ G. Paltrinieri, *La meridiana di San Petronio*, in *L'Astrofilo*, Unione Astrofili Bresciani, bulletin.

⁶ At the Council of Nicaea, 325 AD, to which the Gregorian Reform refers, 1582, it was decided that the sunday following the first full moon after the spring equinox would be Easter.

⁷ R. Buonanno, *Il cielo sopra Roma, i luoghi dell'astronomia*, Springer, Milano, 2008, p. 100.

⁸ The sundial is visible to the Abbey of Cava, located in front of the library, and bears the signature of Rocco Bovi near the *centrum erectum*.

⁹ R. Bovi, *Spiegazione del presente disegno*, Firenze, X agosto 1769: "[...] devotus numini majestatique ejus Rochus Bovius Scyllensis".

¹⁰ N. Severino, *La linea meridiana della Certosa di San Martino a Napoli*, Rocassecca, 2007, in *Gnomonica Italiana* n°23, VIII, 2011.

¹¹ N. Spinosa, *San Martino immagini e memorie*, Grimaldi & C. Editori, 2000, p. 19.

¹² L. Arbace, F. Capobianco, R. Pastorelli, *Il quarto del Priore* Museo della Certosa di San Martino, Sergio Civita Editore, 1986, p. 7.

¹³ Several studies have tried to compare the lists of artworks belonging in the past to the Charterhouse. The first important study was done after the earthquake of 1980 by Teodoro Fittipaldi, who writes in the journal *Christian Art*.

¹⁴ G. Donatone, *Pavimenti e rivestimenti maiolicati in Campania*, Isveimer, 1981, p. 58.

¹⁵ A. Pezzullo, *Relazione tecnica sui lavori nel Quarto del Priore*, Biblioteca Bruno Molajoli a Castel Sant'Elmo a Napoli, 25 maggio 1995.

¹⁶ L. Arbace, F. Capobianco, R. Pastorelli, *op. cit.*, p. 7.

¹⁷ T. Fittipaldi, R. Casolaro, *Il museo di San Martino a Napoli*, Electa Napoli, 1995, p. 61.

¹⁸ G. Donatone, *op. cit.*

¹⁹ G. Donatone, *Maioliche napoletane del '700*, Grimaldi & C. Editori, 2006, p. 37.

²⁰ G. Donatone, *Pavimenti e rivestimenti maiolicati in Campania*, cit., p. 59.

²¹ G. Donatone, *Maiolica delle Due Sicilie*, Edizioni Scientifiche Italiane, 1998, pp. 32-34.

²² The aberration of planets will deal more in detail in Chapter 2, paragraph 2.3.2 of this book.

²³ R. Bovi, *op. cit.*

²⁴ A. Coppola, *Orologi solari e meridiane a Napoli*, Arte Tipografica, 2002, p. 216

²⁵ Nicola Severino, *op. cit.* p. 6.

²⁶ From the definition of Treccani "in measuring instruments, type of metric scale, attributed to Tycho (Brahe, Tynghe), in which each degree is divided into equal parts, and in which we come to assess the tenth part of the range of gradation by a simple procedure of linear interpolation in eye".

²⁷ R. Bovi, *op. cit.*

²⁸ *Magazzino Toscano*, Tomo IV, Parte I, Firenze, MDCCLXXIII, p.131.

²⁹ R. Bovi, *op. cit.*

³⁰ V. paragraph 3.5 of this book.

³¹ R. Bovi, *op. cit.*

³² A. Coppola, *op. cit.*, p. 216

³³ N. Severino, *op. cit.* p.2

³⁴ V. paragraph 3.5 of this book.

³⁵ R. Bovi, *op. cit.*

³⁶ *Magazzino Toscano*, cit., pp. 132-133.

³⁷ More details in paragraph 2.3.2 of this book.

³⁸ Because of the precession of the equinoxes the constellations in which there is nowadays the Sun, on the dates indicated, are no longer the ones to which we refer today in our calendar and to which referred Rocco Bovi, but, being transferred of one constellation to the west they changed toward the Previous one Zodiactal constellation. So, for example, at the spring equinox the Sun is no longer in the constellation of Aries, but in Pisces.

³⁹ N. Severino, *op. cit.*

⁴⁰ V. chapter 2, paragrafo 2.1 of this book.

⁴¹ *ivi*, pp. 43-44.

⁴² The study on geographic locations will be discussed in more detail in Chapter 2, paragraph 2.3.1 of this book.

⁴³ *Magazzino Toscano*, Tomo IV, Parte I, Firenze, MDCCLXXIII, pp. 133.

⁴⁴ *ivi*, pp. 133-134.

⁴⁵ The definition of Treccani Dictionary of Physics: "in the computation of astronomical time, the number which expresses the surplus days of the solar year compared to the moon, usually calculated at January 1st, from which we can immediately ob-

tain all the new moons and the lunar phases of the year (thus serves for the calculation of Easter and the other moveable feasts) "

⁴⁶ The word Epact derives from the greek Epactos (overtaken). Indicates the days that you have to add to the lunar year to get to the solar ore. The epact be distinguished in Annual, monthly and daily. In the case of Rocco Bovi's meridian line we talk about a daily epact. The epacts according to the rule of the Gregorian perpetual calendar is situated in a retrograde order compared to the days of the month, so that the asterisk (note that for example in the 24th day of the month of September) that corresponds to the epact number XXX, which in its turn corresponds the full moon, is in correspondence to January 1, so the epacts XXIX appear in correspondence of day 2 January and so on ... as the lunar month has a duration of between 29 and 30 days, while the civilian can be constituted by 30 or 31 days, except February, it follows that the age of the Moon on each day of the month, that is the average age on the same day of the previous month, plus one.

Month Epact

Genuary 0	February 1	March 0
April 1	May 2	June 3
July 4	Agoust 5	September 7
October 8	November 9	December 10

The epacts XXV and XXIV are put together (as we see for example in correspondence of February 5 on the sundial) so that they correspond to the same day in six different months. Furthermore, next to the epact XXV or XXVI, there is the number 25 written in Arabic numeral, for example, in correspondence dated February the 4st. This happens when the golden number exceeds 11, and the year in question belongs epact XXV, then 25 is the new moon for that year. Both the golden number either the annual epact were readable and well-known thanks to the Tabulae Paschalis which were drawn up following the Reformation. Note the epact concerning a given year to determine the age of the moon on any date of the year is enough to calculate: $1 = (\text{day} + \text{epact mounth} + \text{epatta year})$. Cf. V. de Lucia, *Elementi di Cosmografia e di Cronologia*, II cap. II p. 98, 1831, Fondo Pizzofalcone, Napoli.

⁴⁷ The lunation is the period during which the Moon goes through its phases. Ithe synodic lunar

month which is about 29.25 days is divided into the following phases: new moon (0 days), first quarter (7.5 days), full moon (15 days), last quarter (22.5 days) and moon new or full moon (29.25 days).

⁴⁸ Cf. chapter 2, paragraph 2.3.4 of this book.

⁴⁹ Cf. paragraph 3.5 of this book.

⁵⁰ *ibidem*.

⁵¹ In the technical report on the work directed by the Adele Pezzullo we read that the frescoes on the vaults of the Library and the floor of the Chiaiese have been restored.

⁵² A. Coppola, *op. cit.*, p. 213.

⁵³ N. Severino, *op.cit.*

⁵⁴ *Magazzino Toscano*, *cit.*, p. 130.

⁵⁵ Colomboni A. M., *Prattica gnomonica, o vero, Tauole, con le quali ciascuno ageuolmente può far da sè gli Horologi da Sole orizzontali, verticali e riflessi di qualsiuoglia grandezza*, Bologna 1669, pp. 635-636.

⁵⁶ Figatelli G. M., *Retta linea gnomonica di Giuseppe Maria Figatelli centes. Ouero, breuissima estensione instruttiua frà due punti di chiarezza e facilità per delineare Horologi Horizontali, Verticali e Riflessi, con alcuni pochi de' più vsati frà portatili. Operetta assai diletteuole per il nuouo & vniuersal modo d'operare in qual si sia Horologio Riflesso....*, Viuiano Soliani Stampator Ducale, Modena 1675, p. 51.